

Stars Dataset Analysis

Sebastian Uribe

2026-06-15

Stars Dataset

The [Stars Dataset](#) is a dataset that consists of several features of stars. The goal of the dataset is to verify that stars are able to be represented by a diagram, specifically the [Hertzprung-Russell Diagram](#).

In this document, we will use exploratory data analysis, descriptive statistics, and visualizations to breakdown the features in the dataset to validate the purpose of the dataset.

Glimpse into the Stars Dataset

```
## Rows: 240
## Columns: 7
## $ Temperature..K.      <int> 3068, 3042, 2600, 2800, 1939, 2840, 2637, 2600, ~
## $ Luminosity.L.Lo.     <dbl> 0.002400, 0.000500, 0.000300, 0.000200, 0.00013~
## $ Radius.R.Ro.        <dbl> 0.17000, 0.15420, 0.10200, 0.16000, 0.10300, 0.~
## $ Absolute.magnitude.Mv. <dbl> 16.120, 16.600, 18.700, 16.650, 20.060, 16.980, ~
## $ Star.type            <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, ~
## $ Star.color            <chr> "Red", "Red", "Red", "Red", "Red", "Red", "Red", "Red"~
## $ Spectral.Class       <chr> "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M"~
```

Here we see that the dataset consists of seven features of stars:

1. Temperature (Kelvin)
2. Luminosity (L/Lo)
3. Radius (R/Ro)
4. Absolute Magnitude (Mv)
5. Star Type
6. Star Color
7. Spectral Class

Where:

Lo = 3.828×10^{26} Watts (Avg Luminosity of Sun)

Ro = 6.9551×10^8 m (Avg Radius of Sun)

Let's take a closer look at the elements in the Star Type, Star Color, and Spectral Class columns to understand the visual categories of the stars.

Distinct Star Types, Star Colors, and Spectral Classes

Star Types

```
##      Star.type
## 1          0
## 2          1
## 3          2
## 4          3
## 5          4
## 6          5
```

The distinct elements in the Star.Type column is a range of integers from 0 to 5. The details in the csv file from the Kaggle website gives us a list of what the integers represent:

Brown Dwarf -> Star Type = 0

Red Dwarf -> Star Type = 1

White Dwarf-> Star Type = 2

Main Sequence -> Star Type = 3

Supergiant -> Star Type = 4

Hypergiant -> Star Type = 5

Star Color

```
##      Star.color
## 1          Red
## 2      Blue White
## 3          White
## 4      Yellowish White
## 5      Blue white
## 6  Pale yellow orange
## 7          Blue
## 8      Blue-white
## 9          Whitish
## 10     yellow-white
## 11          Orange
## 12     White-Yellow
## 13          white
## 14          Blue
## 15      yellowish
## 16      Yellowish
## 17      Orange-Red
## 18     Blue white
## 19     Blue-White
```

We see that the Star.color column raises a labeling inconsistency. There are several colors that are repeated due to capitalization, hyphen, and spacing differences. The prominent example is the blue-white color: “Blue White”, “Blue white”, “Blue-white”, and “Blue-White.”

Spectral Classes

```
## Spectral.Class
## 1          M
## 2          B
## 3          A
## 4          F
## 5          O
## 6          K
## 7          G
```

The values in the Spectral.Class column are clean and consistent.

The Spectral Class is a [Stellar Classification](#) that categorizes stars under the Harvard Spectral Classification system using the letters O, B, A, F, G, K, M arranged from hottest (O-Class) to coldest (M-Class).

Each letter class has their own color which provides a better baseline for assigning a color to each star. Therefore, in future visualizations, The Spectral.Class column will be used instead of the Star.color column.

Additionally, the spectral class has levels to each class with O-Class being the highest (hottest) and M-Class being the lowest (coolest).

```
stars <- stars %>%
  mutate(Spectral.Class = factor(Spectral.Class,
                                levels = c("O", "B", "A", "F", "G", "K", "M")))
```

Now we have a better understanding of each column in the dataset:

1. Temperature (Kelvin)
2. Luminosity (L/Lo)
3. Radius (R/Ro)
4. Absolute Magnitude (Mv)
5. Star Type (Red Dwarf, Brown Dwarf, White Dwarf, Main Sequence , SuperGiants, HyperGiants)
6. Star Color (white,Red,Blue,Yellow,yellow-orange etc)
7. Spectral Class (O,B,A,F,G,K,M)

Stars Analysis

Let's take a closer look at the data by using descriptive statistics. Specifically, let's look at the Temperature, Luminosity, and Radius columns of all the stars in the dataset.

Temperature

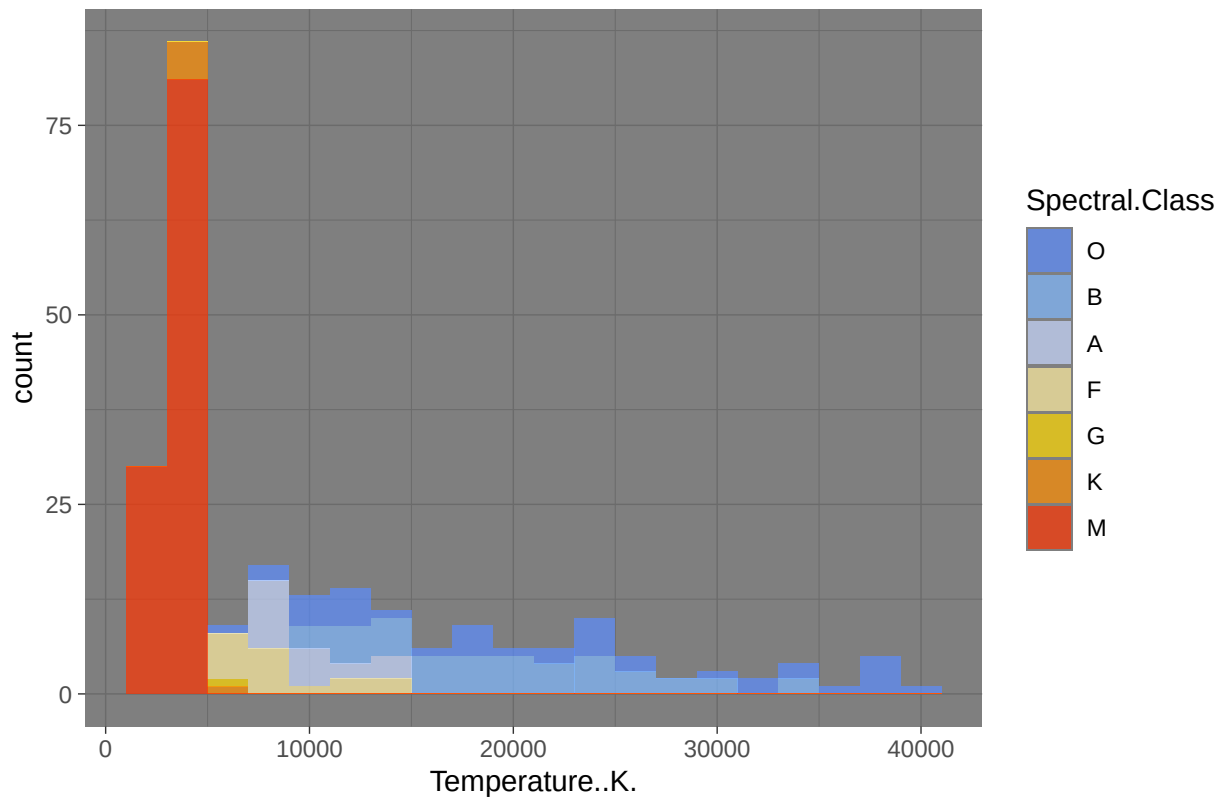
A stars temperature is primarily measured by its color. This is done through [blackbody radiation](#) where the temperature of a star is determined by the wavelengths of light they emit. We should see a trend of red stars being cool where blue/white are the hottest.

```
## 'data.frame': 240 obs. of 3 variables:
## $ Temperature..K.: int 3068 3042 2600 2800 1939 2840 2637 2600 2650 2700 ...
## $ Spectral.Class : Factor w/ 7 levels "O","B","A","F",...: 7 7 7 7 7 7 7 7 7 7 ...
## $ Star.type : int 0 0 0 0 0 0 0 0 0 0 ...
```

```
##      Temperature..K. Spectral.Class Star.type
## 235          21904           B         5
## 236          38940           O         5
## 237          30839           O         5
## 238           8829           A         5
## 239           9235           A         5
## 240          37882           O         5
```

To understand the range of temperatures, let's make a histogram of the temperatures to help us visualize the values.

Distribution of Star Temperature



We see that histogram is right skewed. The majority of the stars have a temperature of less than 10,000K and that many of them are M-Class. The histogram shows a trend that when the temperature increases, their class increases.

```
## # A tibble: 7 x 5
##   Spectral.Class count temp_mean temp_median temp_sd
##   <fct>         <int>    <dbl>      <dbl>    <dbl>
## 1 O             40  22294.    22369  10590.
## 2 B             46  19574.    18850   6195.
## 3 A             19   9842.     9030   1914.
## 4 F             17   8517.     7230   3041.
## 5 G              1   6850      6850    NA
## 6 K              6   4500.     4406.   462.
## 7 M            111   3257.     3324   352.
```

The table tells us the mean, median, standard deviation, and interquartile range for the stars' temperature

per spectral class. From the mean and median columns, the temperature decrease as the spectral classes decrease. From the standard deviation column, the temperatures in the higher classes are highly spread out, which is apparent from the histogram.

G-Class is the smallest class in the data set with only 1 star. The largest class is M-Class which we could also see in the histogram.

Luminosity

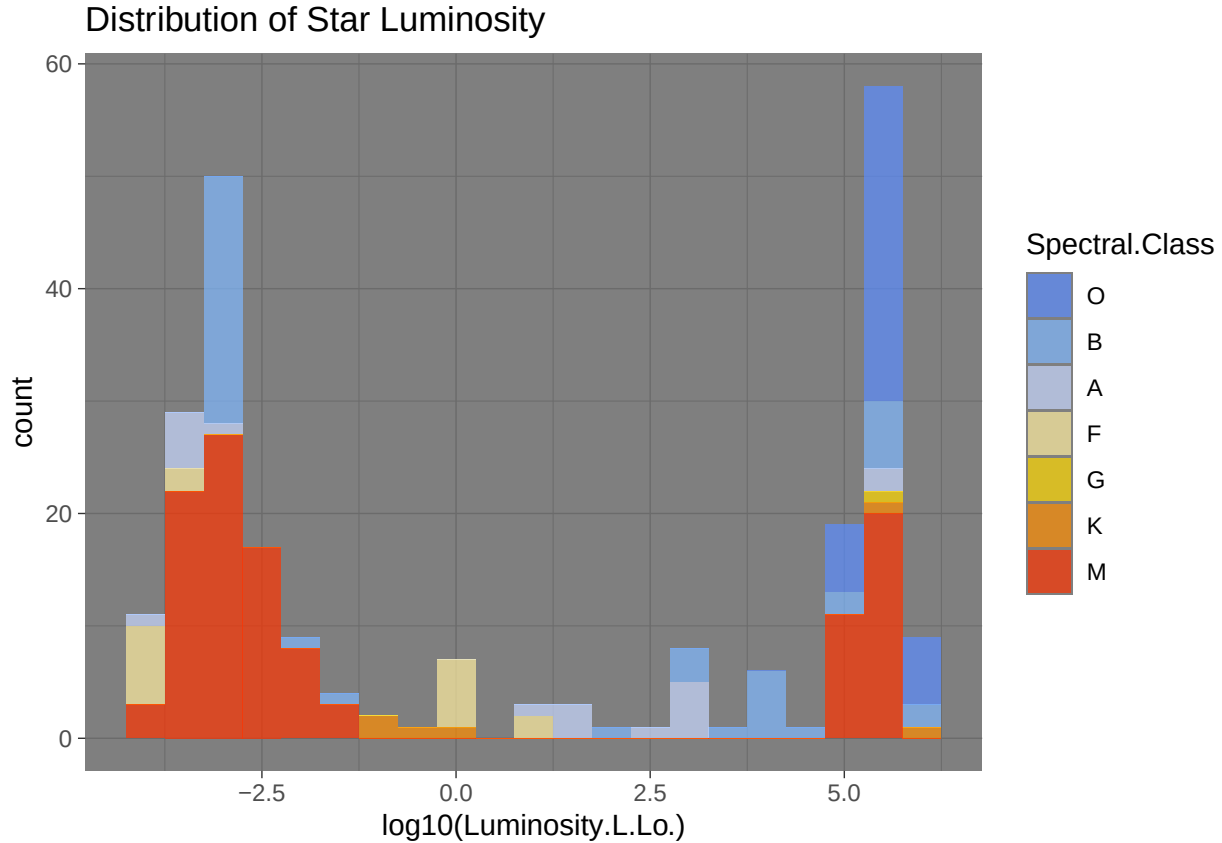
The luminosity of a star measures its total energy output per unit of time. In this dataset, the luminosity of each star is expressed relative to the [Sun's luminosity](#) ($L_o = 3.828 \times 10^{26}$ Watts). For example, a value of $L/L_o = 10$ means the star emits 10 times more energy than our Sun.

```
## 'data.frame': 240 obs. of 3 variables:
## $ Luminosity.L.Lo.: num 0.0024 0.0005 0.0003 0.0002 0.000138 0.00065 0.00073 0.0004 0.00069 0.0001...
## $ Spectral.Class : Factor w/ 7 levels "O","B","A","F",...: 7 7 7 7 7 7 7 7 7 7 ...
## $ Star.type : int 0 0 0 0 0 0 0 0 0 0 ...

## Luminosity.L.Lo. Spectral.Class Star.type
## 235 748490 B 5
## 236 374830 O 5
## 237 834042 O 5
## 238 537493 A 5
## 239 404940 A 5
## 240 294903 O 5
```

By taking a look into the luminosity values, the range goes from ~0.0001 to ~800,000 luminosity. To account for this, the visualizations and descriptive statistics for luminosity will be on a log10 scale.

Let's plot the luminosity values on a histogram.



The histogram follows a bimodal distribution where there are two distinct peaks with a gap between them. The left peak is dominated by M-Class stars which are the brown and red dwarfs. The A-Class and B-Class stars in the left peak are white dwarfs. The gap represents the main sequence stars which is notably sparse in this dataset. The right peak are the supergiants and hypergiants. We see that some M-Class and K-Class stars shine as bright as O-Class and B-Class stars.

Any star is in any class type is able to shine extremely dim to extremely bright.

```
## # A tibble: 7 x 5
##   Spectral.Class count lumi_mean_log10 lumi_median_log10 lumi_sd_log10
##   <fct>         <int>         <dbl>         <dbl>         <dbl>
## 1 O              40           5.46           5.39           0.231
## 2 B              46           0.639          -1.57           3.79
## 3 A              19           0.558           1.58           3.31
## 4 F              17          -1.94          -3.54           2.14
## 5 G               1           5.36           5.36           NA
## 6 K               6           1.45          -0.324           3.25
## 7 M             111          -0.628          -2.62           3.74
```

From the table, the sporadic luminosity of stars regardless of their class is made evident. The noticeable differences in mean and median luminosity show how the extreme cases weigh heavily in these calculations.

Radius

A star's radius is calculated in a similar manner to luminosity. The radius of each star is expressed relative to the Sun's radius ($R_o = 6.9551 \times 10^8$ m). For example, $R/R_o = 10$ means that the star is 10 times larger

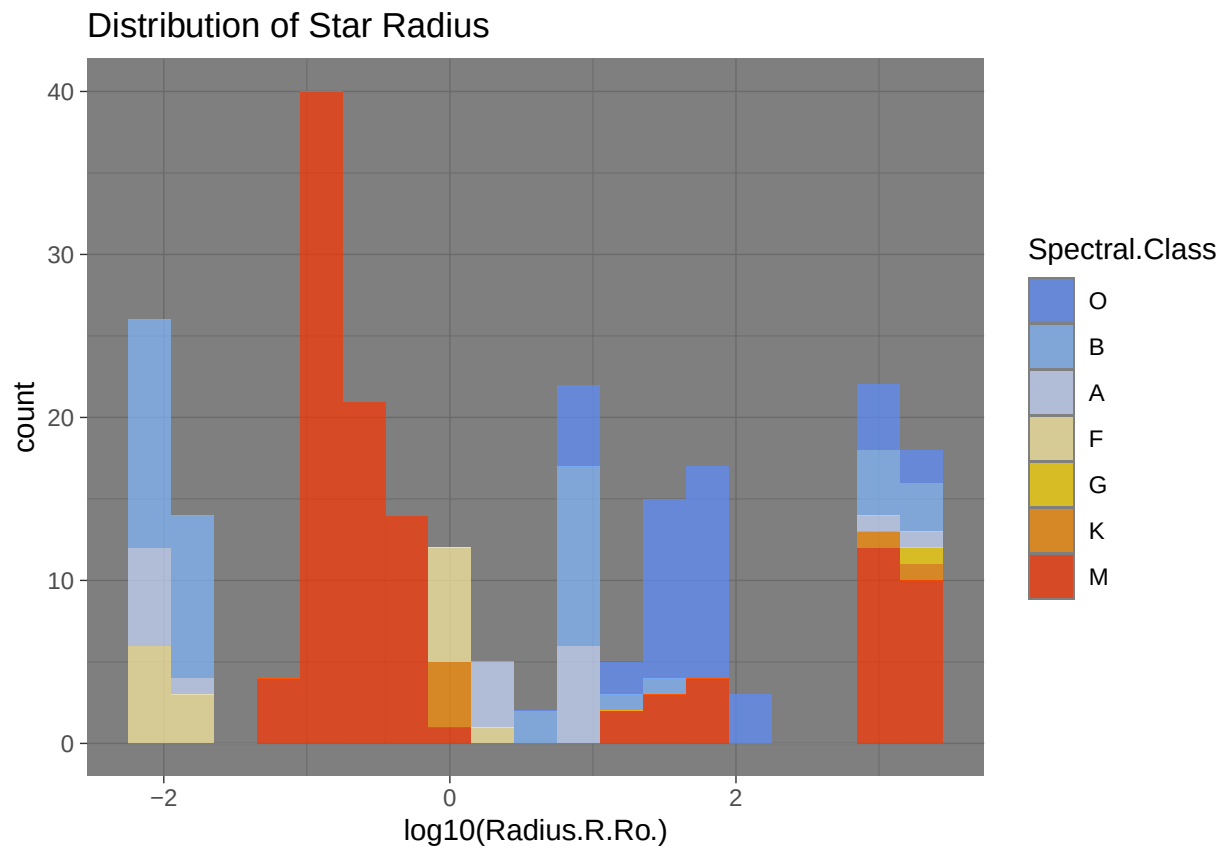
than our Sun.

```
## 'data.frame': 240 obs. of 3 variables:
## $ Radius.R.Ro. : num 0.17 0.154 0.102 0.16 0.103 ...
## $ Spectral.Class: Factor w/ 7 levels "O","B","A","F",...: 7 7 7 7 7 7 7 7 7 7 ...
## $ Star.type : int 0 0 0 0 0 0 0 0 0 0 ...
```

```
##      Radius.R.Ro. Spectral.Class Star.type
## 235          1130              B          5
## 236          1356              O          5
## 237          1194              O          5
## 238          1423              A          5
## 239          1112              A          5
## 240          1783              O          5
```

The radius of stars range from being a small fraction of the size of our Sun to being up to a thousand times larger. We will apply a $\log_{10}$ scale on the radius as well for better interpretation.

Let's plot the radius of stars on a histogram to visualize the data.



The spread of the radii follow a multimodal distribution. The far left cluster are white dwarfs. Despite their compact size, white dwarfs reach very high surface temperatures which gives them their blue-white spectral classification. The middle clusters are the brown dwarfs, red dwarfs, and main sequence stars. These types of stars have similar size to our Sun. The far right cluster are the supergiant and hypergiant stars. We see how M-class stars can reach the size of some of the largest stars in the universe despite their light and temperature output.

```
## # A tibble: 7 x 5
##   Spectral.Class count radius_mean_log10 radius_median_log10 radius_sd_log10
##   <fct>          <int>          <dbl>          <dbl>          <dbl>
## 1 O              40            1.80            1.76            0.666
## 2 B              46           -0.284           -1.84            1.93
## 3 A              19           -0.0885          0.396            1.69
## 4 F              17           -1.02           -1.89            1.04
## 5 G               1            3.17            3.17            NA
## 6 K               6            1.02           -0.00221          1.65
## 7 M            111            0.218           -0.536            1.60
```

The table tells a similar story to the histogram. Comparing the mean, median, and standard deviation radii show how classes like M-Class, F-Class, and A-Class have a diversity of sizes. Taking a look at the mean/median for the G-Class star of 3.17 (log10), this means that the star has a radius of ~1,480 R/R_o. This makes the star a hypergiant, which is rare for this class.

```
##   Temperature..K. Luminosity.L.Lo. Radius.R.Ro. Star.type Spectral.Class
## 1              6850             229000         1467          5             G
```

Main Sequence Imbalance?

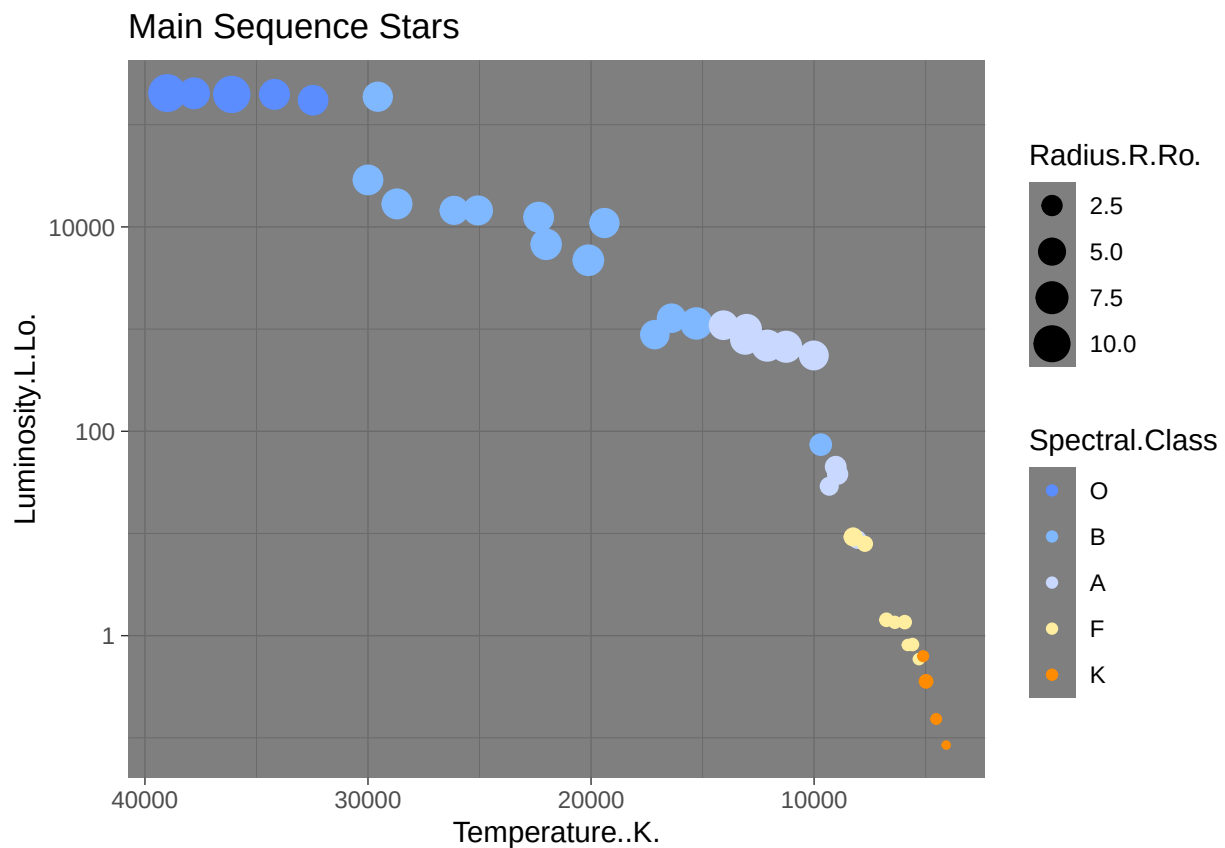
```
## 'data.frame': 40 obs. of 7 variables:
## $ Temperature..K. : int 39000 30000 15276 9700 5800 8052 6757 6380 5936 5587 ...
## $ Luminosity.L.Lo. : num 2.04e+05 2.88e+04 1.14e+03 7.40e+01 8.10e-01 ...
## $ Radius.R.Ro. : num 10.6 6.3 7.2 2.89 0.9 ...
## $ Absolute.magnitude.Mv.: num -4.7 -4.2 -1.97 0.16 5.05 2.42 2.41 2.93 4.46 5.03 ...
## $ Star.type : int 3 3 3 3 3 3 3 3 3 3 ...
## $ Star.color : chr "Blue" "Blue-white" "Blue-white" "Whitish" ...
## $ Spectral.Class : Factor w/ 7 levels "O","B","A","F",...: 1 2 2 2 4 3 4 4 4 4 ...
```

According to [NASA](#), main sequence stars make up about 90% of the stars in the observable universe. The [Harvard Spectral Classification](#) demonstrates the variety of main sequence stars, spanning the entire spectral class.

However in this dataset as mentioned before, main sequence stars make up about 17% of the dataset with 40 of them in total.

The reasoning for this is that the dataset was created for machine learning purposes. The stars in the dataset went through balanced sampling to train machine learning models. Having main sequence stars dominate would bias a model into always predicting this star type.

Let's take a look at the assortment of main sequence stars in the dataset by making a scatter plot comparing the temperature and luminosity, similar to structure of the [HR-Diagram](#).



The diagonal trend of the main sequence stars is present in the scatter plot similar to the one in the HR-Diagram. The upper left is stars with high temperature and high luminosity and the bottom right is

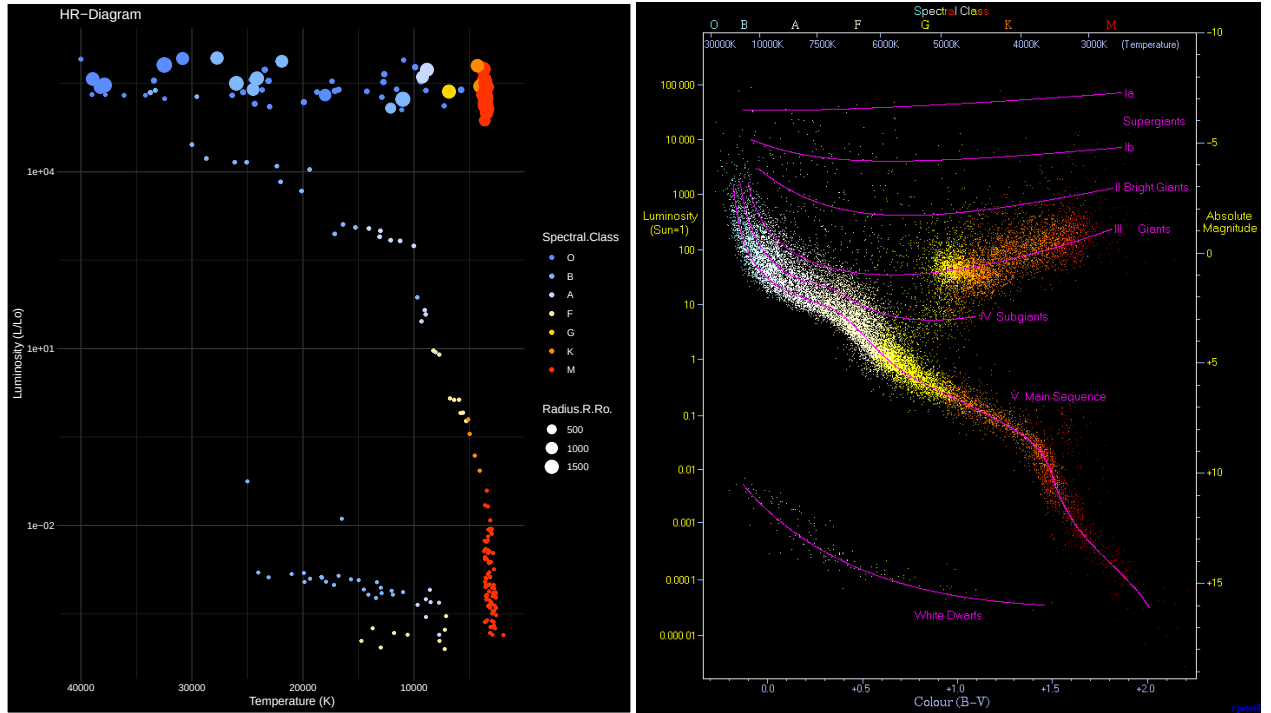
stars with low temperature and low luminosity. We also see that the radius of the stars increases as their temperature and luminosity increases along the diagonal.

M-Class stars are not present in this case despite their [prominence in the main sequence](#). This is due to the limitation of the dataset in `Star.type == 3` excluding red dwarfs.

O-Class stars are present despite their [scarcity in the main sequence](#). This could be explained by the dataset being intended for machine learning or could have been a labeling issue.

HR-Diagram

Finally, let's represent the dataset as an HR-Diagram and compare it to full-fledged one.



Our HR-Diagram demonstrates the main sequence going across the plot as in the actual HR-Diagram. Now we clearly see the M-Class stars on the tail of the diagonal. The group of white dwarfs also similar to the well-established model.

Where the two diagrams differ is in the distribution of the supergiant and hypergiant stars. The plot we created has these stars almost in a line across the top. The actual diagram has the giants clustered and scattered throughout the top of the graph.

Conclusion

Although this dataset presents limitations and possible labeling issues, the dataset still fulfills its purpose: stars follow a pattern, specifically the pattern a HR-Diagram establishes.